# CHIRAG KULAR

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LinkedIn | GitHub | Portfolio | U.S. Permanent Resident

## PROFESSIONAL SUMMARY

**Senior Full-Stack Software Engineer with 8+ years of production experience building and supporting web applications, APIs, data platforms, and distributed workflows.** Combines Python and TypeScript backend engineering with React/Next.js product development, PostgreSQL and MongoDB data modeling, and AWS event-driven systems. Experienced in system integration, performance testing, observability, production debugging, and translating complex workflows into reliable software. Works across architecture, implementation, automated testing, code review, and production support.

## CORE TECHNICAL SKILLS

**Full-Stack:** Python, TypeScript, Node.js, React, Next.js, Angular, Java, REST APIs, application state management
**Databases:** PostgreSQL, MongoDB, SQL Server
**Cloud & Observability:** AWS Lambda, S3, API Gateway, Step Functions, SQS, EKS/Kubernetes, EC2, Glue, CloudWatch, Docker, Terraform, Datadog, Splunk
**Tools & Frameworks:** Lit, Storybook, Playwright, JUnit, GitHub Copilot, Claude, Codex, RAG, OpenAI APIs, AWS Comprehend Medical
**Healthcare & Interoperability:** EHR/EMR, FHIR R4, US Core, HL7v2, Firely Server, Rhapsody Integration Engine, clinical workflows

## PROFESSIONAL EXPERIENCE

### Senior Software Engineer - Full Stack | ICF — Feb 2026 - Present

- Own full-stack modernization of an NCI biospecimen research platform, migrating legacy capabilities to Next.js, TypeScript, PostgreSQL, and AWS EC2 while improving page performance by 40%.
- Translate researcher and operational workflows into accessible, data-driven product features, partnering with product, design, engineering, and domain stakeholders from requirements through production.
- Use Datadog APM and distributed tracing to diagnose service behavior and improve production reliability and performance visibility.

### Software Engineer - Contract | M&T Bank (TEKsystems) — Oct 2024 - Feb 2026

- Designed a modular TypeScript/Lit component platform for the Currency design system, with 30+ reusable components adopted across 10+ product teams, establishing consistent APIs, accessibility patterns, and Storybook documentation.
- Built automated scaffolding and Playwright coverage for shared components, accelerating delivery and reducing regression risk across consuming applications.
- Partnered with UX, design, and application teams to turn complex workflows into accessible interfaces; improved maintainability through reviews, standards, and documented patterns.

### Senior Software Engineer - Backend | ICF — Jul 2023 - Oct 2024

- Built secure AWS microservices that ingest, validate, process, and export healthcare data for the CMS Innovation Center's EOM and GUIDE models using Node.js/Python, Lambda, S3, SQS, Step Functions, API Gateway, MongoDB, and EKS.
- Designed asynchronous REST APIs for FHIR submissions and status polling; integrated Firely Server validation and persisted workflow state and metadata for reliable, traceable processing.
- Implemented event-driven workflows for file intake, ClamAV malware scanning, eligibility and metadata validation, failure recovery, reprocessing, and result export.
- Performance-tested 150K FHIR resources and a 250K-record end-to-end workload; tuned Lambda concurrency, throttling, batching, and Step Functions distributed-map behavior.
- Evaluated EHRbase, FHIRPath rules, US Core R4 profiles, and provenance/history patterns to support interoperability and long-term data correctness.
- Created Splunk and CloudWatch dashboards for load balancers, Lambda, MongoDB, workflow duration, health, and errors; independently diagnosed pipeline and production issues.
- Owned services from technical design through operational readiness; reviewed code and provided architecture, testing, and maintainability guidance across the team.

### Software Engineer | MatrixCare — Aug 2018 - Jun 2023

- Built and maintained clinician-facing workflows in MatrixCare's skilled-nursing EHR using Java/Spring MVC, TypeScript/Angular, REST APIs, SQL Server, and AWS.
- Delivered Medication Reconciliation and Orders enhancements in CareAssist that streamlined patient readmissions and improved workflow efficiency by 50%.
- Integrated LabSoft laboratory data with the SNF platform, added HL7 fields, and updated Rhapsody Integration Engine routes to improve clinical interoperability and data reliability.
- Designed and optimized T-SQL stored procedures with execution-plan analysis for clinical workflows, reporting, and SAP Crystal Reports.
- Built reusable Angular forms, services, directives, and pipes for complex EHR workflows; partnered with UX and clinical stakeholders to simplify end-user interactions.
- Applied AWS Comprehend Medical to extract medication frequency from pharmacy instructions and evaluated practical NLP use cases for clinical data.
- Led two interns on an Okta authorization proof of concept using React, Python, and AWS Lambda; led a micro-frontend evaluation for the legacy Java platform.
- Provided on-call production support, diagnosed urgent customer issues, refactored legacy code safely, and expanded JUnit coverage during feature delivery.

**Software Engineer | iGATE (now Capgemini)** **Nov 2013 - Jun 2015**

- Supported Oracle Applications R12 financial modules and PL/SQL-based data conversion, requirements analysis, and end-to-end quality assurance.

## PROJECTS

- HL7-FHIR Mapping Agent (2026): Prototype for mapping synthetic HL7v2 and C-CDA samples into FHIR R4, with deterministic parsing, retrieval-assisted mapping, and bounded validation.
- Scan Tip (2025-2026): Built and published a SwiftUI app for tip calculations, receipt scanning, possible service-charge or included-gratuity alerts, saved dining history, and iCloud sync.

## EDUCATION

**M.S., Computer Science** - Rochester Institute of Technology, 2018 | GPA: 3.46
**B.E., Information Technology** - University of Mumbai, 2013 | GPA: 3.4

## CERTIFICATIONS

- Claude Certified Developer - Foundations
- AWS Certified Data Engineer - Associate
- AWS Certified Solutions Architect - Associate
- AWS Certified Developer - Associate
- Datadog Fundamentals
- Datadog APM & Distributed Tracing Fundamentals
- Microsoft Certified: Azure AI Fundamentals

## PATENT

Systems and Methods for Providing Prompts on an Electronic Device - US-20230004865-A1, published Jan 2023 (patent pending)